# TIMEMIXER++: A General Time Series Pattern Machine for Universal Predictive Analysis

A multi-scale, multi-resolution framework for forecasting, classification, anomaly detection, and imputation

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## Abstract

- Tasks: forecasting, classification, anomaly, imputation.
- Introduce TSPM: TIMEMIXER++.
- Idea: mix across time and frequency.
- Modules: MRTI, TID, MCM, MRM.
- State-of-the-art on eight tasks.

Figure 1 placeholder: Benchmarks and CKA summary Adaptive representations across tasks



## 1. Introduction

- Time series have overlapping periodicities.
- RNN/TCN: weak long-range capture.
- Transformers: struggle with overlap.
- Need multi-scale, multi-period modeling.
- Align with real-world multi-scale generation.

Goal: one model for diverse temporal structures.



## Contributions

- Convert series to multi-resolution time images.
- Dual-axis decomposition: seasonality and trend.
- Mix across scales and resolutions.
- SOTA on 8 tasks, 30 benchmarks.

8 Tasks | 30 Benchmarks | 27 Baselines | SOTA Results

## 2. Related Work

### Time Series Models

- ARIMA, STL: trend/seasonal; limited nonlinearity.
- RNN/TCN: local focus; short dependencies.
- Transformers: long-range via attention.
- TimesNet, MLPs: 2D reshaping and simplicity.

### Hierarchical Modeling

- Prior: moving averages then modeling.
- Limitation: rigid decomposition.
- Ours: dual-axis latent attention.
- Scales and resolutions interact hierarchically.

**Positioning** Flexible latent decomposition

## 3. Architecture

- Encoder-only: input, MixerBlocks, output.
- Input: downsample to multi-scale set.
- MixerBlock: MRTI → TID → MCM → MRM.
- Output: scale heads and ensemble.

Figure 2 placeholder: Framework flow 1D → time images → mix → outputs

**Design** From 1D to rich 2D

## 3. Multi-scale Time Series

- Input $x_0 \in R^{T \times C}$.
- Downsample M times with stride 2.
- $X_{init} = \{x_0 \dots x^M\}$ with shorter lengths.

```
x_m = ConvStride2(x_{m-1})
```

**Scales** Coarse to fine context

## 3.1 Input Projection

- Channel mixing at coarsest scale.
- Variate-wise self-attention across variables.
- Embed all scales to d_model.

```
[Q_M,K_M,V_M]=Linear(x^M); Channel-Attn
```

**Cross-variable** Global integration

## 3.2 MixerBlock

- LayerNorm and residual stack.
- MRTI: FFT-guided period selection.
- TID: dual-axis attention outputs s, t.
- MCM: seasonal bottom-up; trend top-down.
- MRM: amplitude-weighted fusion.

**Block Flow** MRTI → TID → MCM → MRM

## 4. Experiments: Summary

- Eight tasks and thirty benchmarks.
- Consistent SOTA versus twenty-seven baselines.
- Strong generalization and transfer ability.

| | | |
|---|---|---|
| **Long-term**<br>Best MSE/MAE on eight sets. | **M4 uni**<br>Lowest SMAPE, MASE, OWA. | **PeMS multi**<br>Large MAE/MAPE reductions. |
| **Imputation**<br>Lowest errors across masks. | **Few-shot**<br>Wins with 10% training data. | **Zero-shot**<br>Best ETT transfers. |
| **Classify**<br>75.9% accuracy overall. | **Anomaly**<br>F1 87.47% across sets. | **Takeaway**<br>One model wins broadly. |

**Scope** Single-model superiority

## Representation Analysis

- Time images: original, seasonality, trend.
- Multi-period and time-varying trends.
- CKA: task-dependent layer similarity.

Figure 4 placeholder: Time image visualizations Scale- and resolution-aware features

**Insight** Adaptive depth patterns

## 5. Conclusion

- Multi-resolution imaging and dual-axis decomposition.
- Mixing across scales and resolutions.
- SOTA over eight tasks and domains.

See appendices for details and limitations.

**End** Universal pattern machine

# Limitations

- Non-periodic spikes may be under-modeled.
- Sensitivity to top-K frequency selection.
- Compute overhead from 2D conversions.
- Large memory at high resolutions.